Designing new simulation

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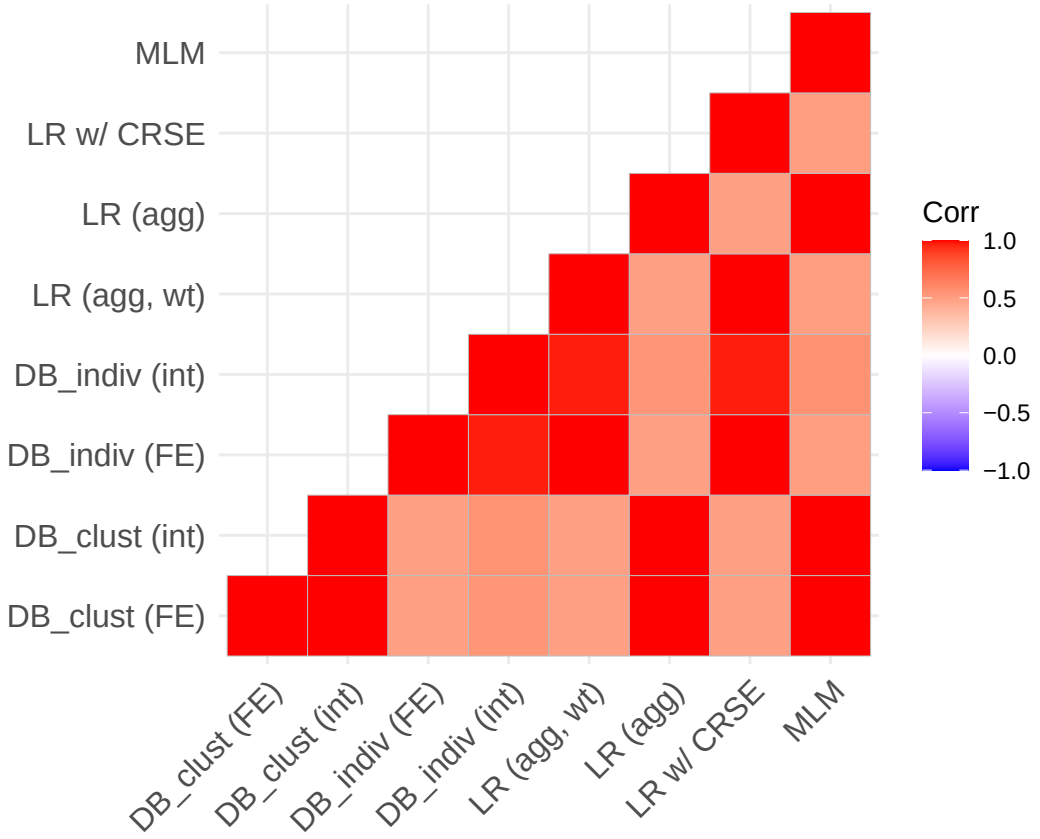
Picking representative models

```
cor_mat <- function(d, value, lab=F) {
  d %>%
    select(ID, method, val={{value}}) %>%
    pivot_wider(names_from=method, values_from=val) %>%
    select(-ID) %>%
    cor() %>%
    ggcorrplot::ggcorrplot(type="lower",
                           lab = lab,
                           show.diag=T)
}

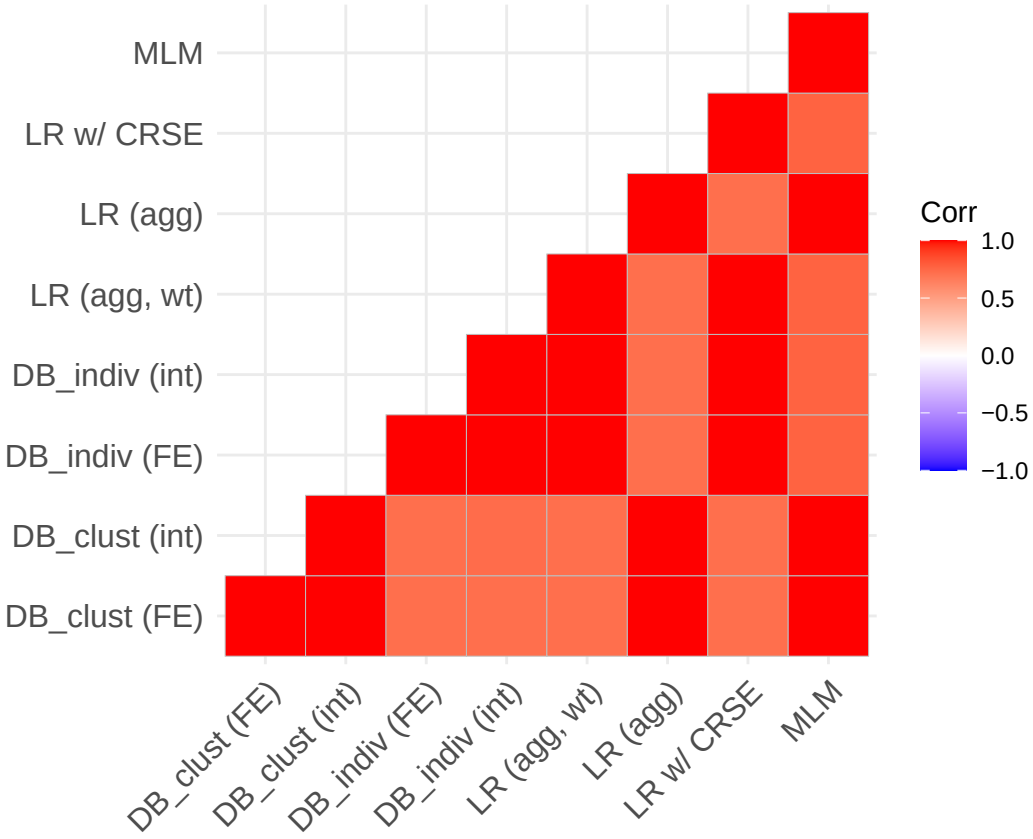
names(sruns)
```

```
##  [1] "ID"                          "method"
##  [3] "K"                           "J.bar"
##  [5] "p.bar"                       "sd.district"
##  [7] "ICC.district"                "variable.J"
##  [9] "variable.p"                  "n.bar"
## [11] "sd.school"                   "ICC.school"
## [13] "variable.n"                  "size.impact.cor.district"
## [15] "size.impact.cor.school"      "proptx.impact.cor.district"
## [17] "ATE.super.district"          "ATE.super.school"
## [19] "ATE.super.student"           "ATE.super.dist.student"
## [21] "ATE.super.dist.school"       "sim.per.min"
## [23] "R"                           "mean.SE.hat"
## [25] "bias.f.stu"                  "bias.f.sch"
## [27] "bias.f.dstu"                 "bias.f.dsch"
## [29] "bias.s.stu"                  "bias.s.sch"
## [31] "bias.s.dstu"                 "bias.s.dsch"
## [33] "RMSE.f.stu"                  "RMSE.f.sch"
## [35] "RMSE.f.dstu"                 "RMSE.f.dsch"
## [37] "RMSE.s.stu"                  "RMSE.s.sch"
## [39] "RMSE.s.dstu"                 "RMSE.s.dsch"
## [41] "SE.s"                        "sd.SE.hat.s"
## [43] "cover.f.stu"                 "cover.f.sch"
## [45] "cover.f.dstu"                "cover.f.dsch"
## [47] "cover.s.stu"                 "cover.s.sch"
## [49] "cover.s.dstu"                "cover.s.dsch"
## [51] "SE.f"                        "sd.SE.hat.f"
## [53] "inflate.f"                   "inflate.s"
```

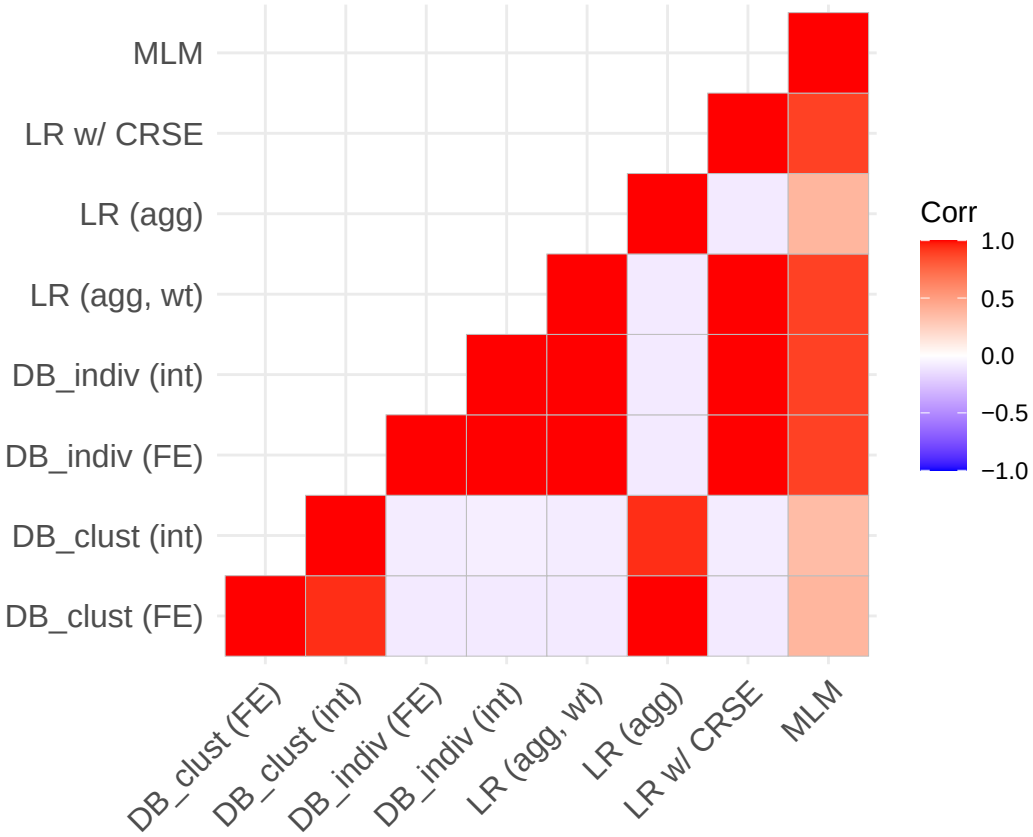
```
cor_mat(sruns, bias.f.stu)
```



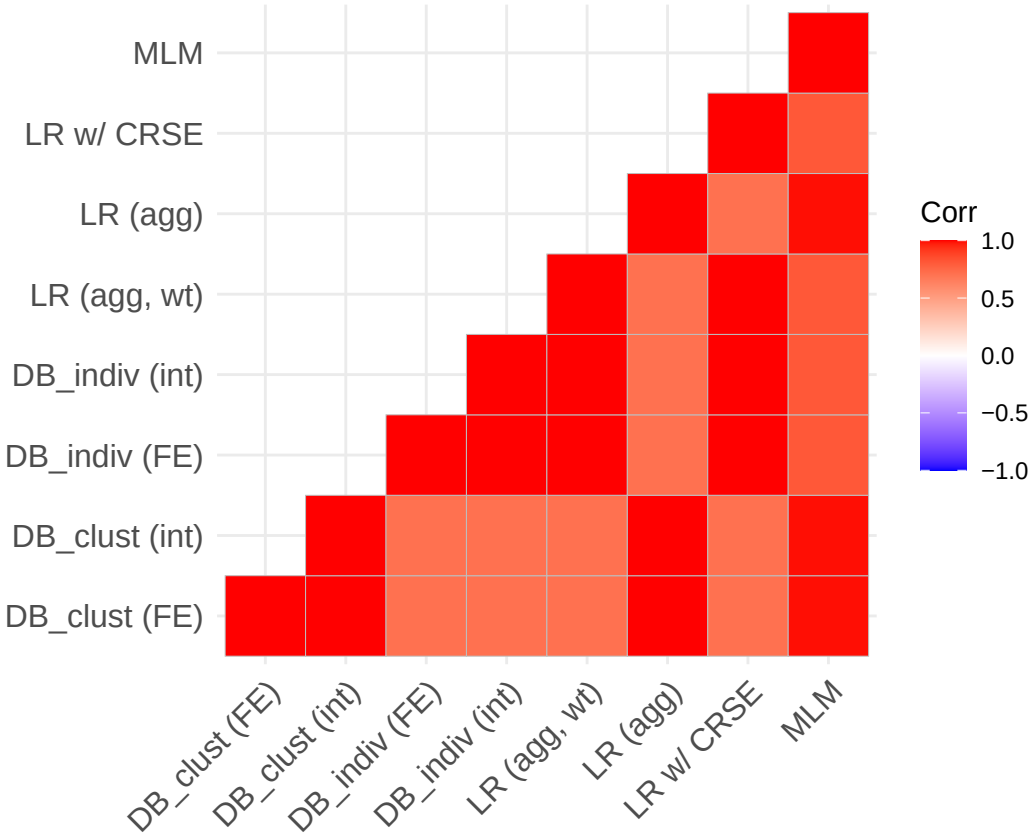

```
cor_mat(sruns, bias.f.dstu)
```



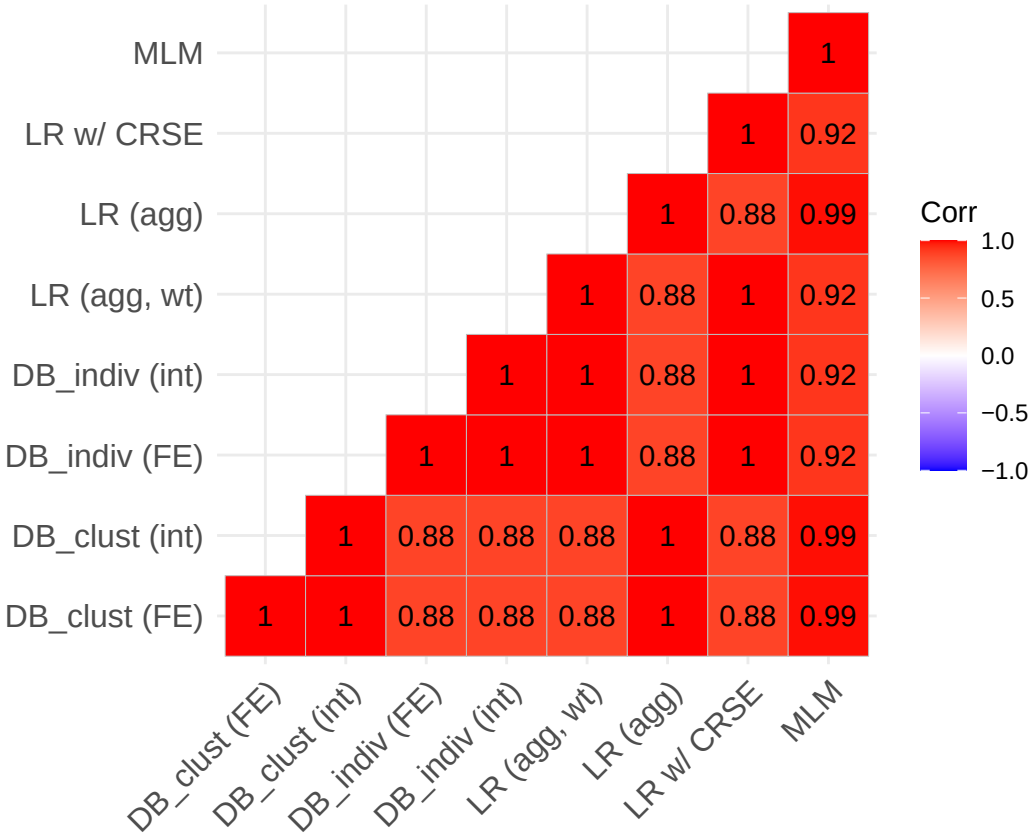

```
cor_mat(sruns, bias.f.sch)
```




```
cor_mat(sruns, bias.f.dsch)
```




```
cor_mat(sruns, RMSE.f.stu, lab=T)
```



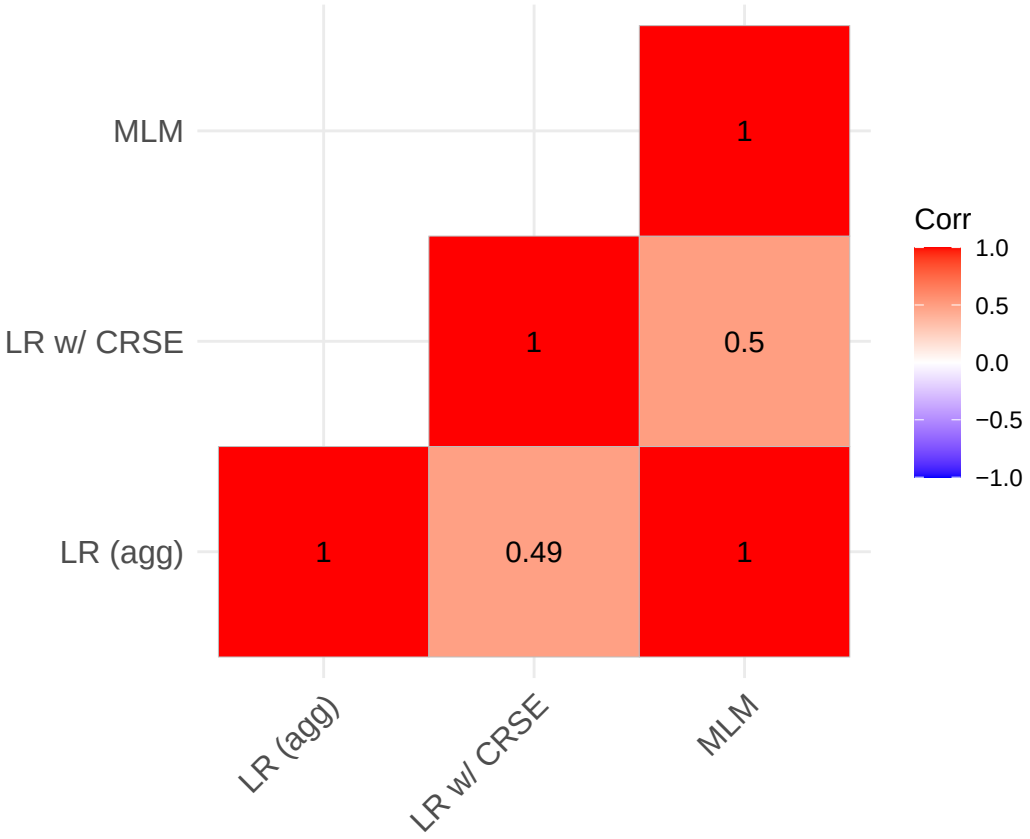
Conclusions:

- LR with CRSE = LR (agg, wt) = DB_indiv (FE) ≈ DB_indiv (int)
- LR (agg) = DB_clust (int) = DB_clust (FE)

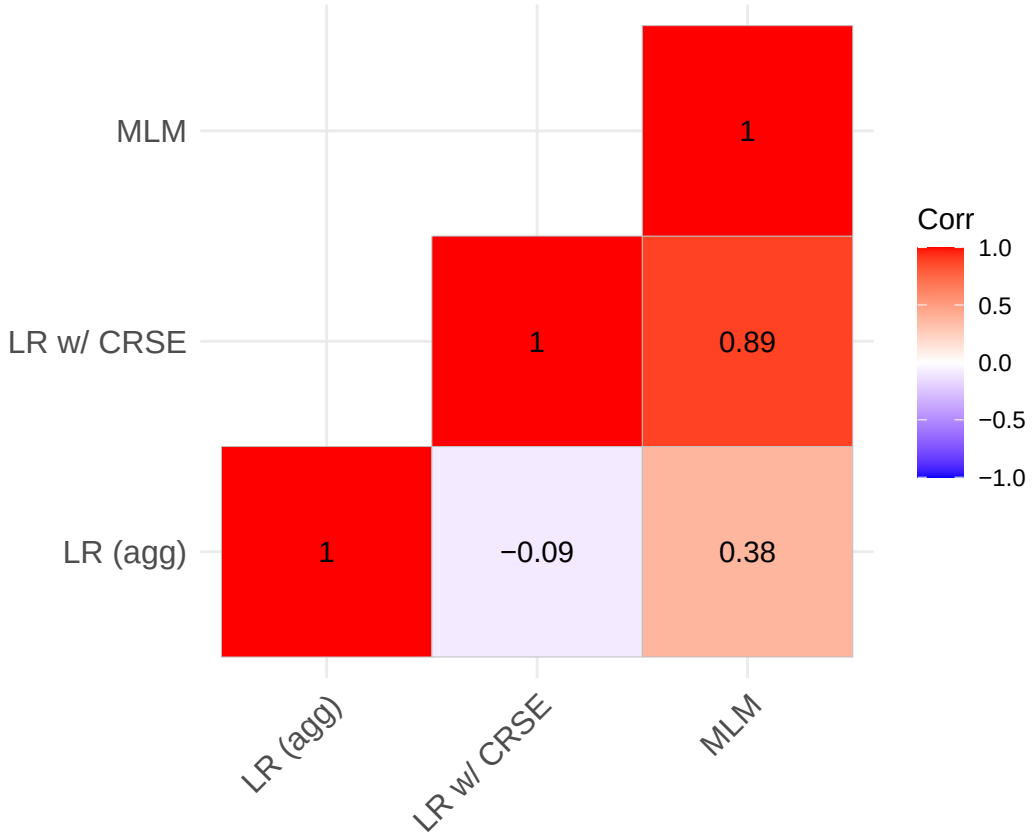
So just use LR with CRSE, LR (agg), and MLM

```
sruns2 <- sruns %>%
  filter(method %in% c("LR w/ CRSE", "LR (agg)", "MLM"))

cor_mat(sruns2, bias.f.stu, lab=T)
```



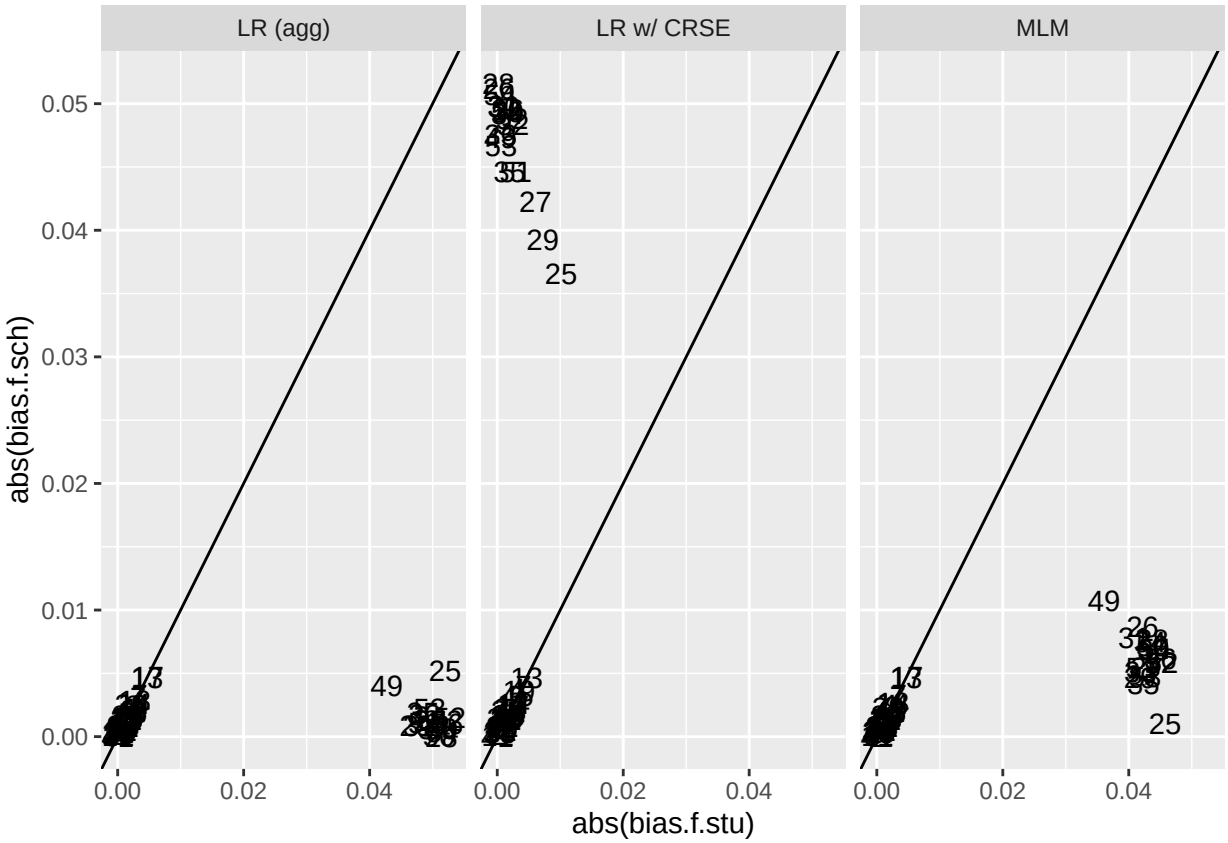

```
cor_mat(sruns2, bias.f.sch, lab=T)
```



Breaking down simulations

Let's identify when models are biased for the non-district-weighted estimands.

```
sruns2 %>%
  select(ID, method, bias.f.stu, bias.f.sch) %>%
  ggplot(aes(x=abs(bias.f.stu), y=abs(bias.f.sch), label=ID)) +
  geom_text() +
  geom_abline() +
  facet_wrap(~method)
```



Let's generate the two clumps. Note that the IDs in the clumps are the same across methods. Also note that J.bar doesn't affect things

```
b_expected <- sruns2 %>%
  filter(abs(bias.f.stu) > 0.02 & method == "LR (agg)") %>%
  pull(ID)

sruns2 %>%
  filter(method == "MLM", J.bar == 50) %>%
  select(ID, all_of(vary_colsJ)) %>%
  mutate(class = case_when(ID %in% b_expected ~ "biased_expected",
                           TRUE ~ "unbiased")) %>%
  arrange(ID)
```

```
##    ID J.bar variable.J variable.n variable.p size.impact.cor.district
## 1   2    50       TRUE       TRUE       TRUE                      0.00
## 2   4    50      FALSE       TRUE       TRUE                      0.00
## 3   6    50       TRUE       TRUE      FALSE                      0.00
## 4   8    50      FALSE       TRUE      FALSE                      0.00
## 5  10    50       TRUE      FALSE       TRUE                      0.00
## 6  12    50      FALSE      FALSE       TRUE                      0.00
## 7  14    50       TRUE      FALSE      FALSE                      0.00
## 8  16    50      FALSE      FALSE      FALSE                      0.00
## 9  18    50       TRUE       TRUE       TRUE                      0.75
## 10 20    50       TRUE       TRUE      FALSE                      0.75
## 11 22    50       TRUE      FALSE       TRUE                      0.75
## 12 24    50       TRUE      FALSE      FALSE                      0.75
```

```
## 13 26    50       TRUE       TRUE       TRUE                    0.00
## 14 28    50      FALSE       TRUE       TRUE                    0.00
## 15 30    50       TRUE       TRUE      FALSE                    0.00
## 16 32    50      FALSE       TRUE      FALSE                    0.00
## 17 34    50       TRUE       TRUE       TRUE                    0.75
## 18 36    50       TRUE       TRUE      FALSE                    0.75
## 19 38    50       TRUE       TRUE       TRUE                    0.00
## 20 40    50      FALSE       TRUE       TRUE                    0.00
## 21 42    50       TRUE      FALSE       TRUE                    0.00
## 22 44    50      FALSE      FALSE       TRUE                    0.00
## 23 46    50       TRUE       TRUE       TRUE                    0.75
## 24 48    50       TRUE      FALSE       TRUE                    0.75
## 25 50    50       TRUE       TRUE       TRUE                    0.00
## 26 52    50      FALSE       TRUE       TRUE                    0.00
## 27 54    50       TRUE       TRUE       TRUE                    0.75
##    size.impact.cor.school proptx.impact.cor.district           class
## 1                       0                       0.00        unbiased
## 2                       0                       0.00        unbiased
## 3                       0                       0.00        unbiased
## 4                       0                       0.00        unbiased
## 5                       0                       0.00        unbiased
## 6                       0                       0.00        unbiased
## 7                       0                       0.00        unbiased
## 8                       0                       0.00        unbiased
## 9                       0                       0.00        unbiased
## 10                      0                       0.00        unbiased
## 11                      0                       0.00        unbiased
## 12                      0                       0.00        unbiased
## 13                      1                       0.00 biased_expected
## 14                      1                       0.00 biased_expected
## 15                      1                       0.00 biased_expected
## 16                      1                       0.00 biased_expected
## 17                      1                       0.00 biased_expected
## 18                      1                       0.00 biased_expected
## 19                      0                       0.75        unbiased
## 20                      0                       0.75        unbiased
## 21                      0                       0.75        unbiased
## 22                      0                       0.75        unbiased
## 23                      0                       0.75        unbiased
## 24                      0                       0.75        unbiased
## 25                      1                       0.75 biased_expected
## 26                      1                       0.75 biased_expected
## 27                      1                       0.75 biased_expected
```

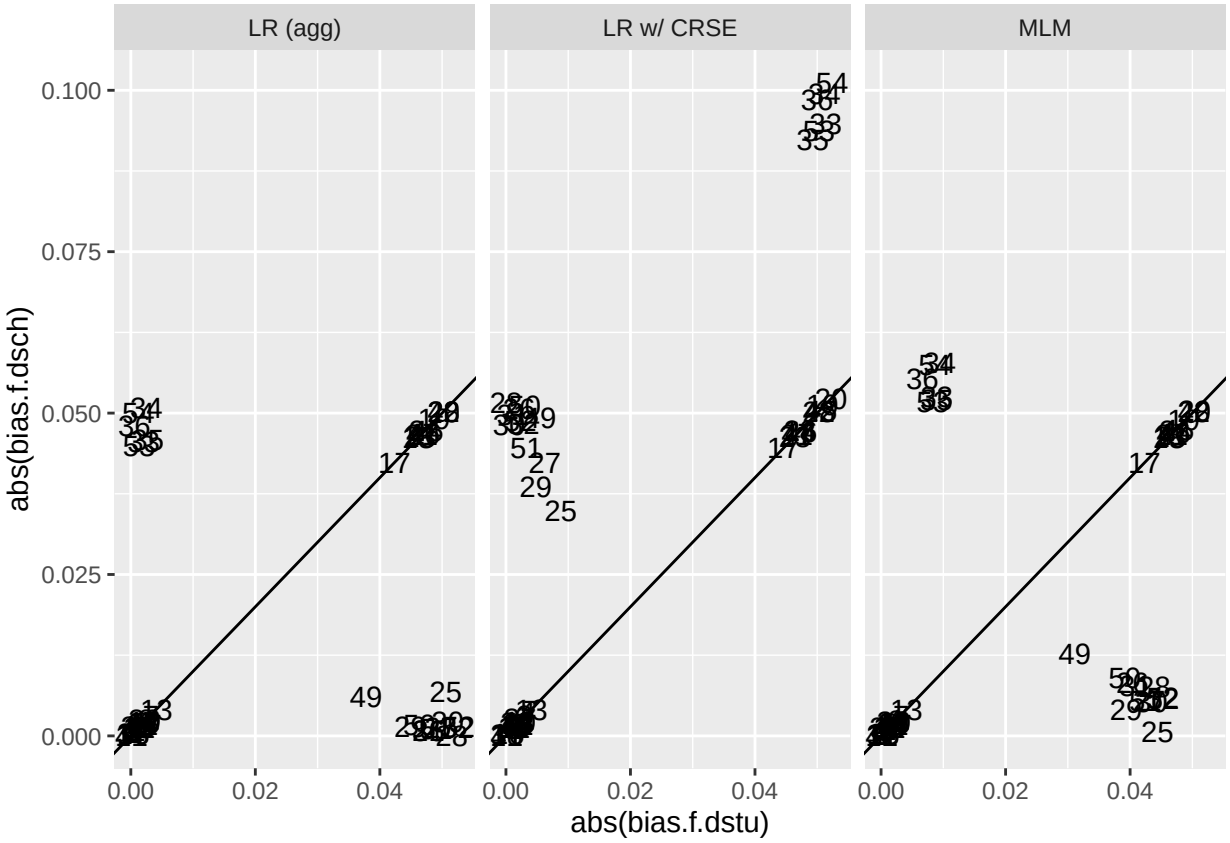
Conclusion: for usual estimands, correlating treatment effects with school sizes induces expected biases.

Same as above, but for district stuff

Now for the district-weighted estimands.

```
sruns2 %>%
  select(ID, method, bias.f.dstu, bias.f.dsch) %>%
  ggplot(aes(x=abs(bias.f.dstu), y=abs(bias.f.dsch), label=ID)) +
  geom_text() +
  geom_abline() +
```

```
  facet_wrap(~method)
```



Weird! Now we get that the methods may be biased for neither, either, or both of the estimands! Let's generate the four clumps. Note that the clumps are the same for all of the methods. Also, J.bar doesn't affect the clumps, so I'll just look at J.bar = 50.

```
b_neither <- sruns %>%
  filter(method == "LR (agg)", abs(bias.f.dstu) < 0.01, abs(bias.f.dsch) < 0.01) %>%
  pull(ID)
b_both <- sruns %>%
  filter(method == "LR (agg)", abs(bias.f.dstu) > 0.02, abs(bias.f.dsch) > 0.02) %>%
  pull(ID)
b_weird <- sruns %>%
  filter(method == "LR (agg)", abs(bias.f.dstu) < 0.01, abs(bias.f.dsch) > 0.02) %>%
  pull(ID)
b_expected <- sruns %>%
  filter(method == "LR (agg)", abs(bias.f.dstu) > 0.02, abs(bias.f.dsch) < 0.01) %>%
  pull(ID)

sruns %>%
  filter(method == "MLM", J.bar == 50) %>%
  select(ID, all_of(vary_colsJ)) %>%
  mutate(class = case_when(ID %in% b_neither ~ "unbiased",
                           ID %in% b_both ~ "biased_both",
                           ID %in% b_expected ~ "biased_expected",
                           ID %in% b_weird ~ "weird"))
```

```
##    ID J.bar variable.J variable.n variable.p size.impact.cor.district
## 1  10    50       TRUE      FALSE       TRUE                     0.00
## 2  12    50      FALSE      FALSE       TRUE                     0.00
## 3  14    50       TRUE      FALSE      FALSE                     0.00
## 4  16    50      FALSE      FALSE      FALSE                     0.00
## 5  18    50       TRUE       TRUE       TRUE                     0.75
## 6   2    50       TRUE       TRUE       TRUE                     0.00
## 7  20    50       TRUE       TRUE      FALSE                     0.75
## 8  22    50       TRUE      FALSE       TRUE                     0.75
## 9  24    50       TRUE      FALSE      FALSE                     0.75
## 10 26    50       TRUE       TRUE       TRUE                     0.00
## 11 28    50      FALSE       TRUE       TRUE                     0.00
## 12 30    50       TRUE       TRUE      FALSE                     0.00
## 13 32    50      FALSE       TRUE      FALSE                     0.00
## 14 34    50       TRUE       TRUE       TRUE                     0.75
## 15 36    50       TRUE       TRUE      FALSE                     0.75
## 16 38    50       TRUE       TRUE       TRUE                     0.00
## 17  4    50      FALSE       TRUE       TRUE                     0.00
## 18 40    50      FALSE       TRUE       TRUE                     0.00
## 19 42    50       TRUE      FALSE       TRUE                     0.00
## 20 44    50      FALSE      FALSE       TRUE                     0.00
## 21 46    50       TRUE       TRUE       TRUE                     0.75
## 22 48    50       TRUE      FALSE       TRUE                     0.75
## 23 50    50       TRUE       TRUE       TRUE                     0.00
## 24 52    50      FALSE       TRUE       TRUE                     0.00
## 25 54    50       TRUE       TRUE       TRUE                     0.75
## 26  6    50       TRUE       TRUE      FALSE                     0.00
## 27  8    50      FALSE       TRUE      FALSE                     0.00
##    size.impact.cor.school proptx.impact.cor.district           class
## 1                       0                       0.00        unbiased
## 2                       0                       0.00        unbiased
## 3                       0                       0.00        unbiased
## 4                       0                       0.00        unbiased
## 5                       0                       0.00     biased_both
## 6                       0                       0.00        unbiased
## 7                       0                       0.00     biased_both
## 8                       0                       0.00     biased_both
## 9                       0                       0.00     biased_both
## 10                      1                       0.00 biased_expected
## 11                      1                       0.00 biased_expected
## 12                      1                       0.00 biased_expected
## 13                      1                       0.00 biased_expected
## 14                      1                       0.00           weird
## 15                      1                       0.00           weird
## 16                      0                       0.75        unbiased
## 17                      0                       0.00        unbiased
## 18                      0                       0.75        unbiased
## 19                      0                       0.75        unbiased
## 20                      0                       0.75        unbiased
## 21                      0                       0.75     biased_both
## 22                      0                       0.75     biased_both
## 23                      1                       0.75 biased_expected
## 24                      1                       0.75 biased_expected
## 25                      1                       0.75           weird
```

```
## 26                      0                      0.00        unbiased
## 27                      0                      0.00        unbiased
```

Conclusions: * Unbiased when: no size-impact correlations at school and district level * Biased as expected when: size-impact correlation at school level but not at district level * Biased for both when: size-impact correlation at district level but not at school level * Weird when: size-impact correlation at both school and district level

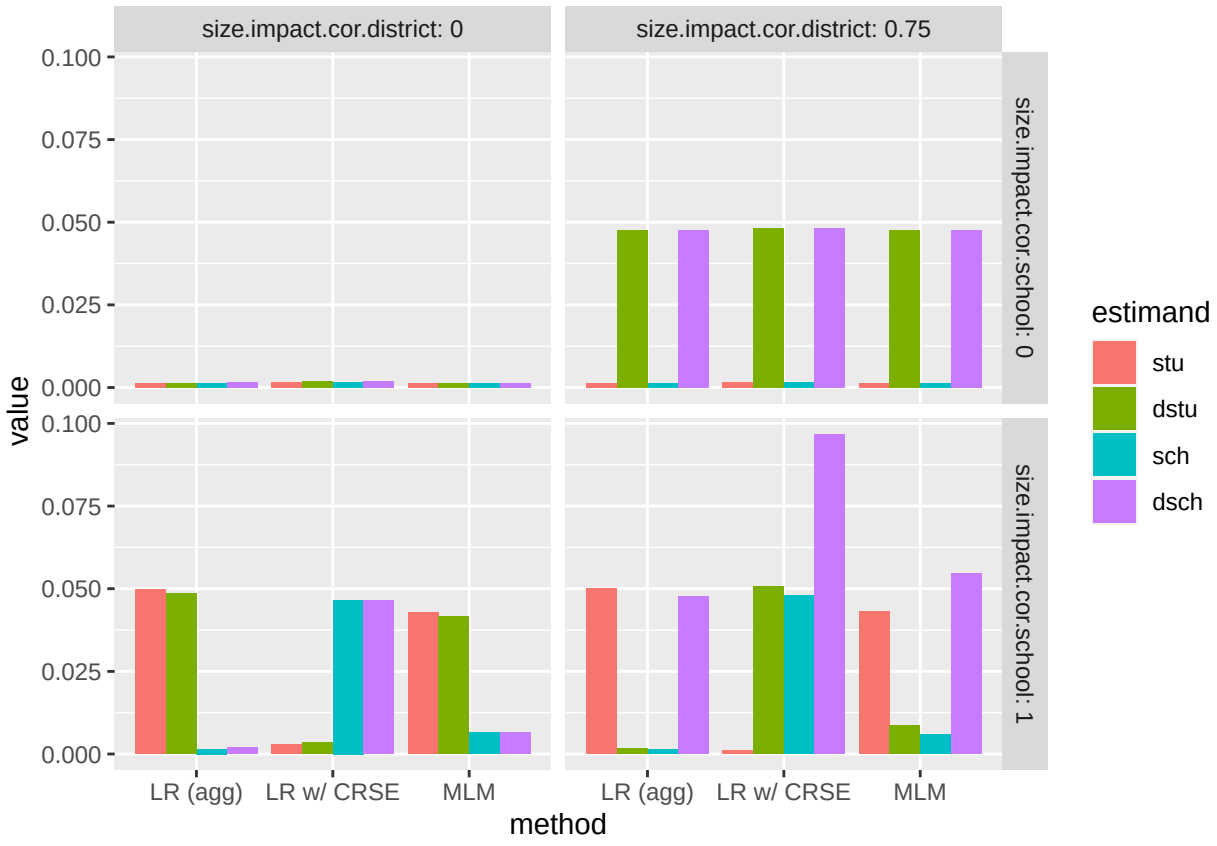
Takeaways: * Only thing that matters is size-impact correlations (i.e., proptx correlations, J.bar don't matter)

Conclusions

Size-impact correlations matter. We can facet by these size-impact correlations.

```
sruns2 %>%
  group_by(method, size.impact.cor.school, size.impact.cor.district) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value, fill=estimand)) +
  geom_col(position="dodge") +
  facet_grid(size.impact.cor.school ~ size.impact.cor.district, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method', 'size.impact.cor.school'. You can
## override using the `.groups` argument.
```

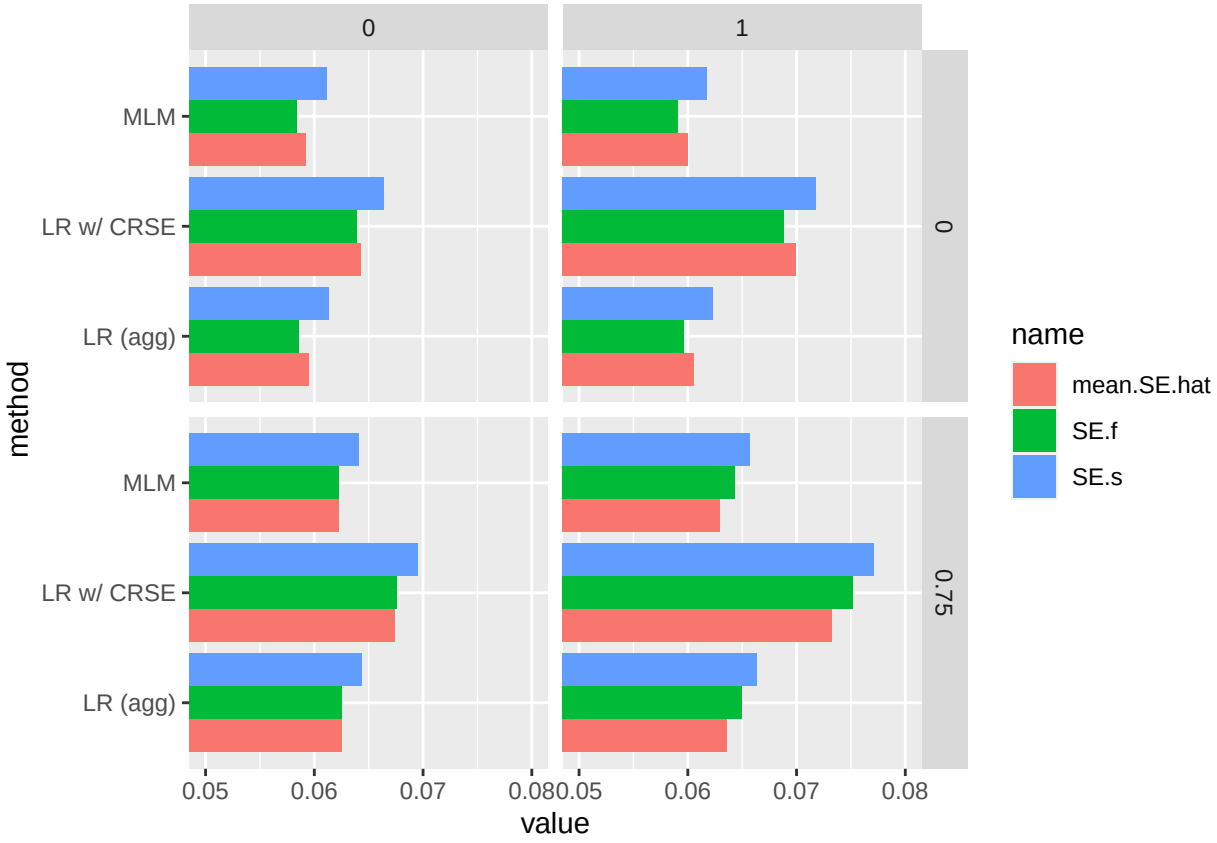


For bias, we see the expected stories: * no bias if no correlations * bias for district-weighted estimands if only district correlations * expected bias for non-district-weighted estimands if only school correlations * chaos if both! expected bias for non-district-weighted estimands; for district-weighted estimands, backwards bias for LR (agg) and MLM, just much more bias than expected for LR w/ CRSE.

Checking: SEs by factor

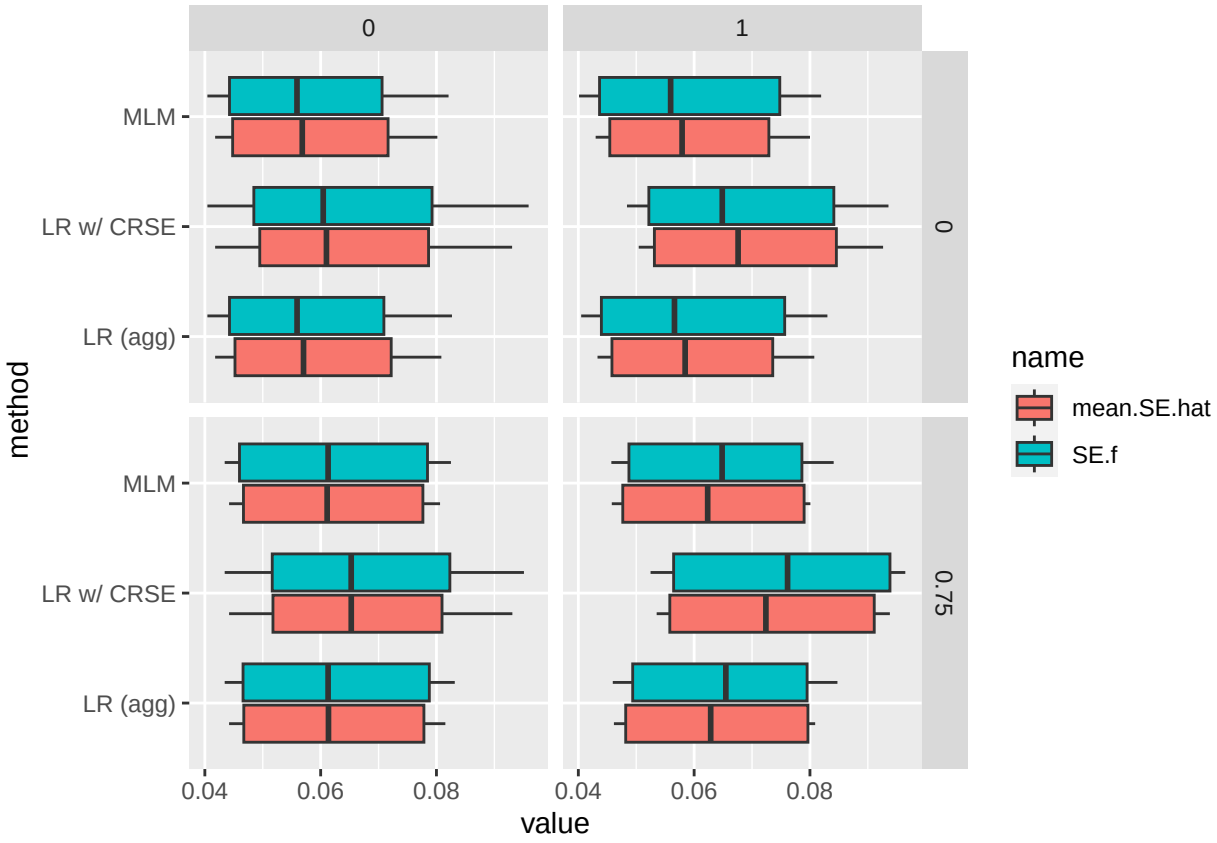
```
sruns %>%
  filter(method %in% key_methods) %>%
  group_by(method, size.impact.cor.district, size.impact.cor.school) %>%
  summarize(across(starts_with("SE."), ~mean(.)),
            mean.SE.hat = mean(mean.SE.hat)) %>%
  pivot_longer(contains("SE")) %>%
  ggplot(aes(x=method, y=value, group=name, fill=name)) +
  geom_col(position="dodge") +
  facet_grid(size.impact.cor.district ~ size.impact.cor.school) +
  coord_flip(ylim = c(0.05, 0.08))
```

```
## `summarise()` has grouped output by 'method', 'size.impact.cor.district'. You
## can override using the `.groups` argument.
```




```
sruns %>%
  filter(method %in% key_methods) %>%
  # pivot_longer(cols = c(SE.f, SE.s, mean.SE.hat)) %>%
  pivot_longer(cols = c(SE.f, mean.SE.hat)) %>%
  ggplot(aes(x=method, y=value, fill=name)) +
```

```
  geom_boxplot() +
  facet_grid(size.impact.cor.district ~ size.impact.cor.school) +
  coord_flip()
```

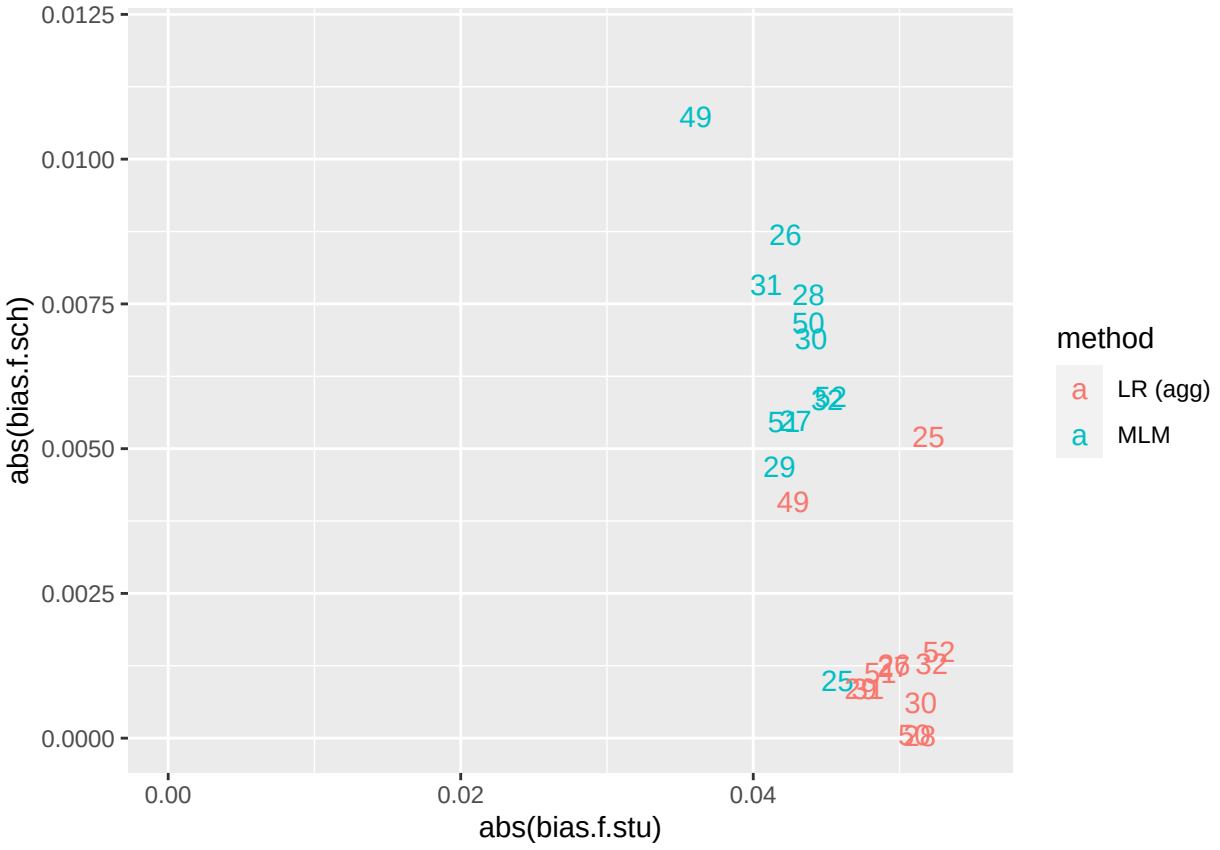


Uhhh... not clear that any of these are actually that far off?

Checking: is MLM always targeting student-weighted?

MLM is pretty consistently worse at sch (which it's good at nonetheless), and better at stu (which it's bad at regardless)

```
sruns %>%
  filter(method %in% c("LR (agg)", "MLM"),
         size.impact.cor.district == 0,
         size.impact.cor.school == 1) %>%
  select(ID, method, starts_with("bias")) %>%
  ggplot(aes(x=abs(bias.f.stu), y=abs(bias.f.sch), color=method, label=ID)) +
  geom_text() +
  coord_cartesian(ylim=c(0,0.012), xlim=c(0,0.055))
```



Could be worth checking scenarios that exacerbate this behavior?

To do in simulation 2

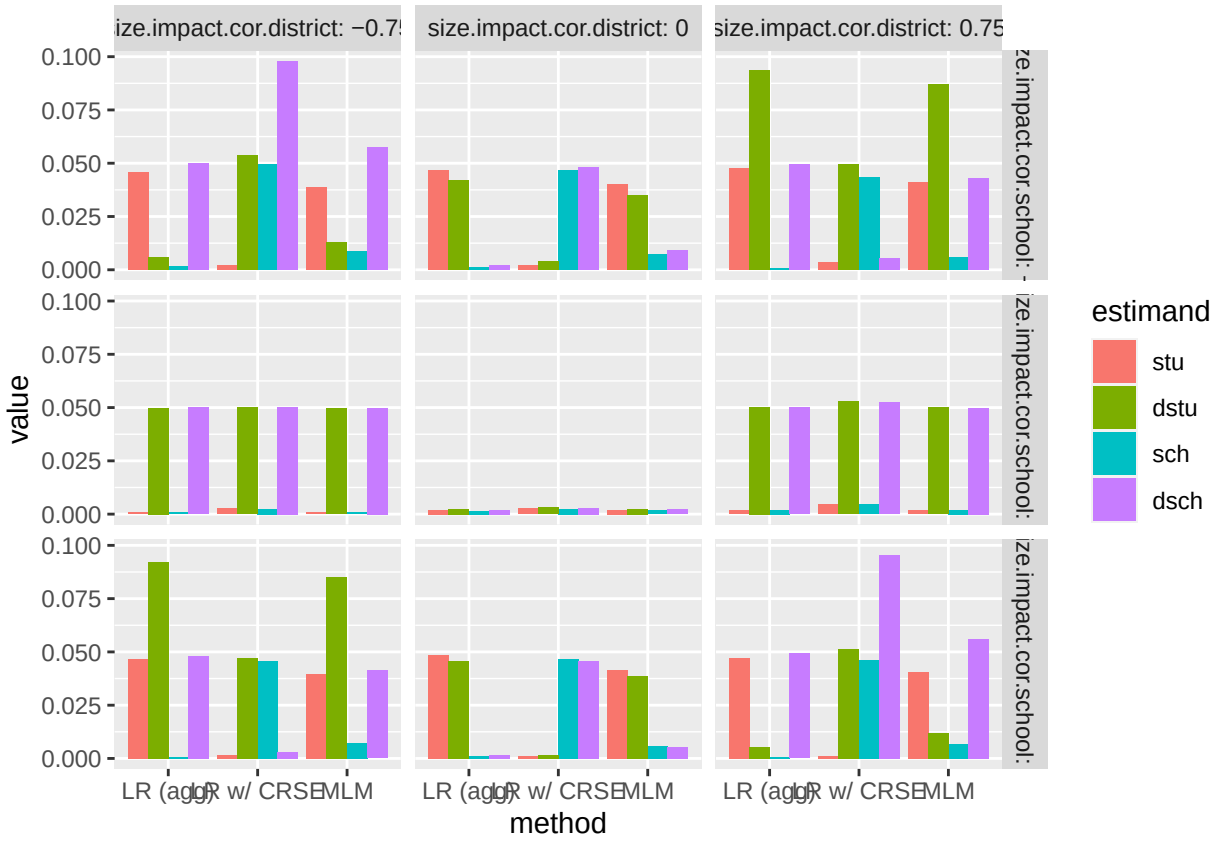
More futzing around with the correlations!

Correlation results

```
load( "~/R/clusterRCTsim/results/aggregated_simulation_results2.RData" )
```

```
sruns %>%
  filter(method %in% key_methods) %>%
  group_by(method, size.impact.cor.school, size.impact.cor.district) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value, fill=estimand)) +
  geom_col(position="dodge") +
  facet_grid(size.impact.cor.school ~ size.impact.cor.district, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method', 'size.impact.cor.school'. You can
## override using the `.groups` argument.
```



Results: * zero school-level correlation: all methods unbiased for non-district-weighted estimands. If district-level correlation is nonzero, all methods biased for district-weighted estimands. * zero district-level correlation: if school-level correlation is nonzero, all methods biased as expected for student/school estimands.

Chaos reigns on the diagonals! LR (agg) and MLM are the same, but… * Same sign correlation: MLM and LR (agg) biased for stu (expected) and dsch (why?), LR w/ CRSE biased for dstu (why?), sch (expected), dsch (expected but magnitude is huge) * Different sign correlation: in reverse! LR w/ CRSE biased for sch (expected) and dstu (why?), other two biased for dsch (why?), stu (expected), dstu (expected but magnitude is huge)

REASON: we're just adding biases together! Clearly the corners of the plot below are just sums of the sides

```
sruns %>%
  filter(method %in% key_methods) %>%
  group_by(method, size.impact.cor.school, size.impact.cor.district) %>%
  summarize(across(starts_with("bias"), mean)) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value, fill=estimand)) +
  geom_col(position="dodge") +
  facet_grid(size.impact.cor.school ~ size.impact.cor.district, labeller = "label_both")
```

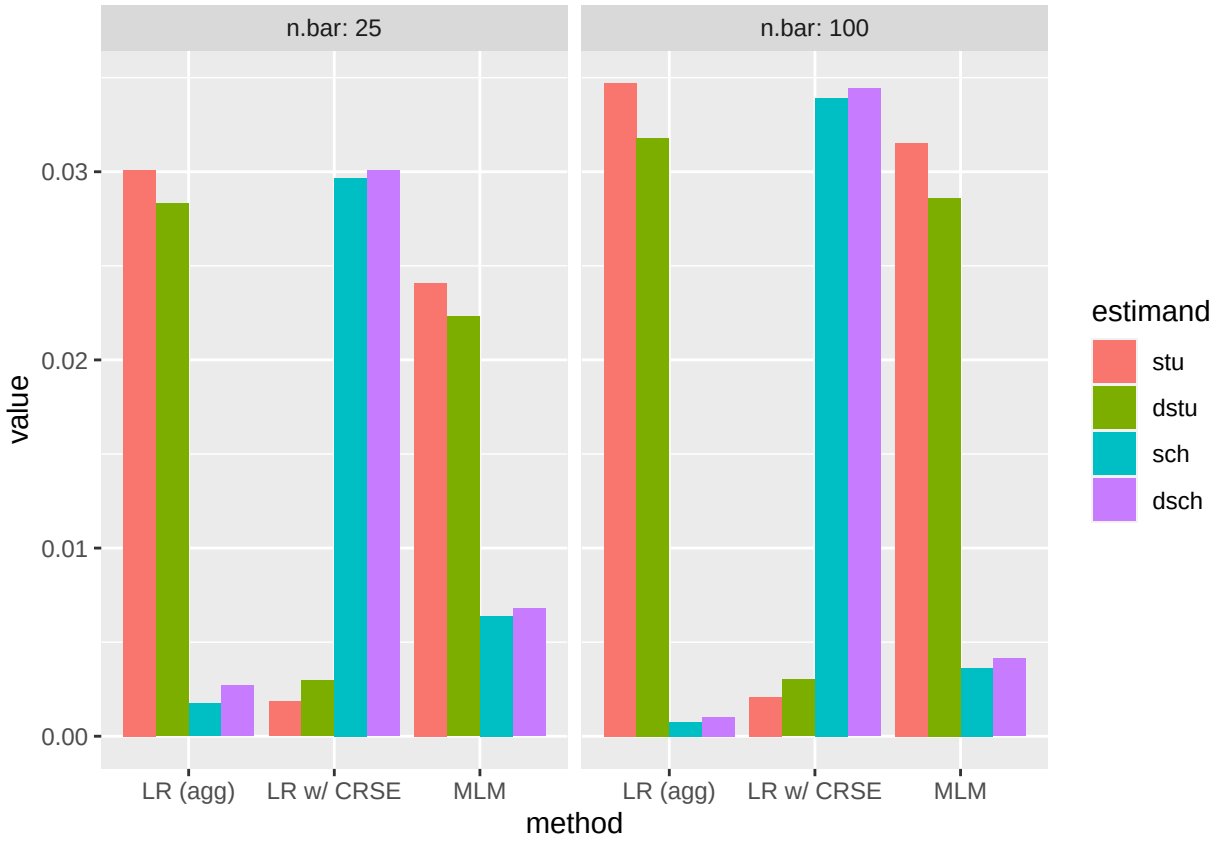
```
## `summarise()` has grouped output by 'method', 'size.impact.cor.school'. You can
## override using the `.groups` argument.
```



Checking sample sizes

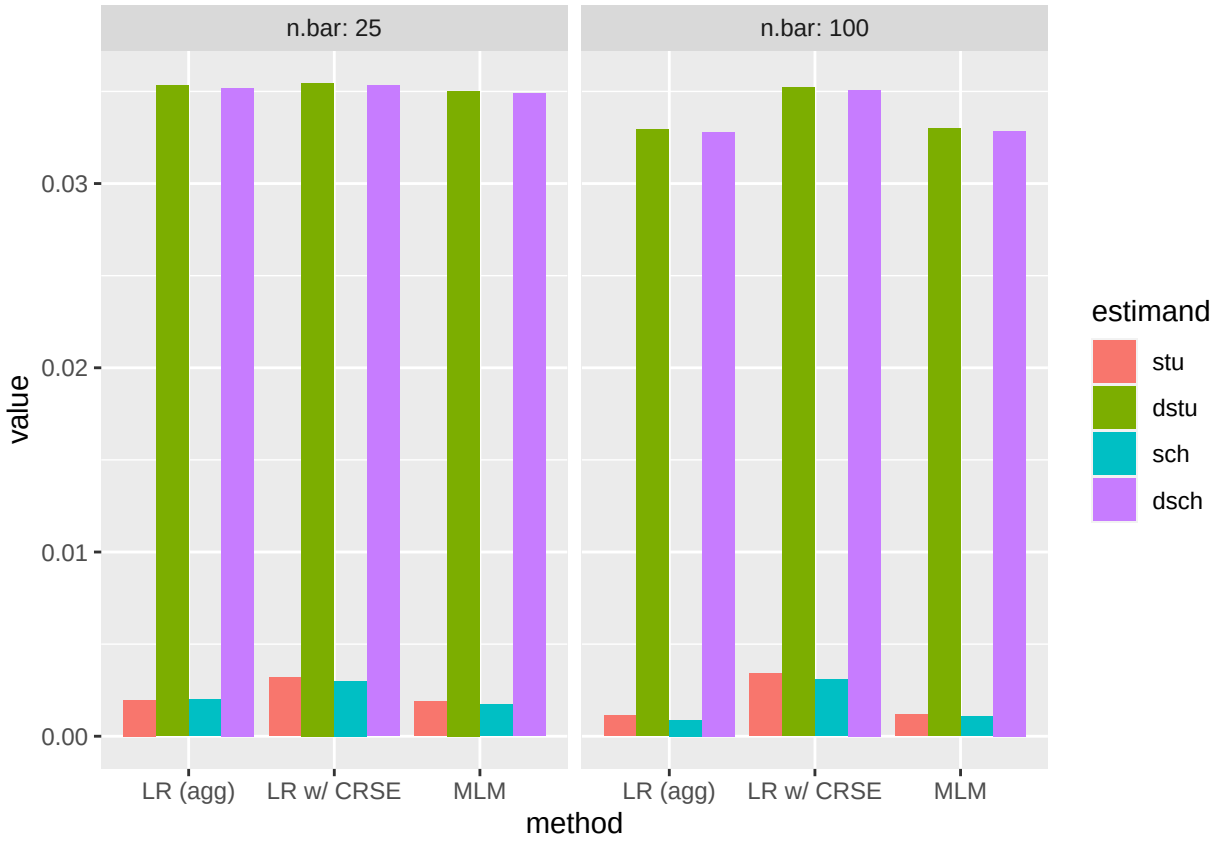
```
sruns %>%
  filter(method %in% key_methods) %>%
  filter(size.impact.cor.district == 0) %>%
  group_by(method, n.bar) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value, fill=estimand)) +
  geom_col(position="dodge") +
  facet_grid( ~ n.bar, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```




```
sruns %>%
  filter(method %in% key_methods) %>%
  filter(size.impact.cor.school == 0) %>%
  group_by(method, n.bar) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value, fill=estimand)) +
  geom_col(position="dodge") +
  facet_grid( ~ n.bar, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```



Might be some effect? Hard to tell...

Conclusion:

Results when both impact correlations are nonzero are confusing because they are the sum of the effects of changing each of the correlations! Sometimes the effects magically cancel out and you're left with no bias!

So:

1. Correlating district impacts with district sizes leads to bias for district-weighted estimands
2. Correlating school impacts with school sizes leads to bias for non-target estimands (both district-weighted and non-district-weighted)

Also, sample size might matter?

To do in simulation 3

Torture the simulation parameters (in terms of data sizes, etc)!

Results

```
load( "~/R/clusterRCTsim/results/aggregated_simulation_results3.RData" )
```

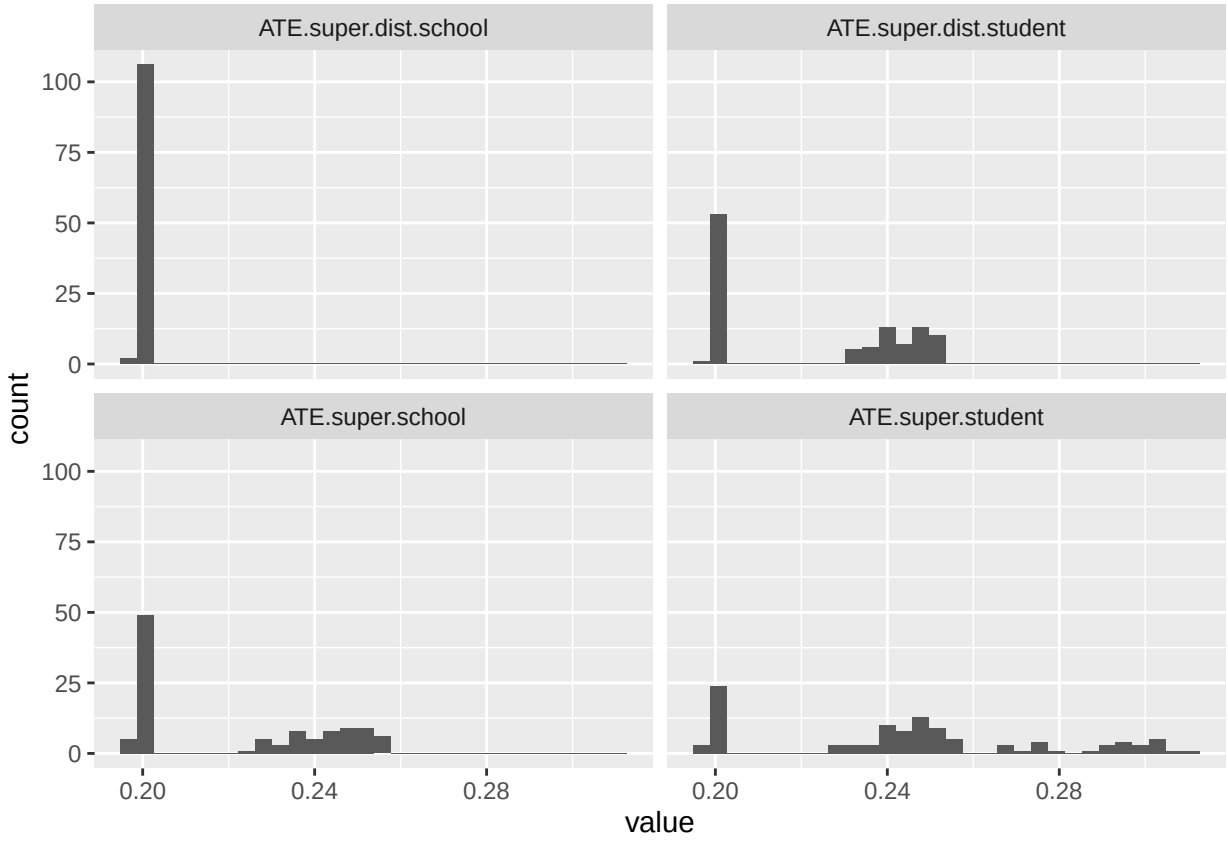
Does anything happen?

Check ranges of true estimand values

```
sruns %>%
  select(-ATE.super.district) %>%
```

```
  group_by(ID) %>%
  summarize(across(starts_with("ATE."), first)) %>%
  pivot_longer(starts_with("ATE.")) %>%
  ggplot(aes(x=value)) +
  geom_histogram() +
  facet_wrap(~name)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



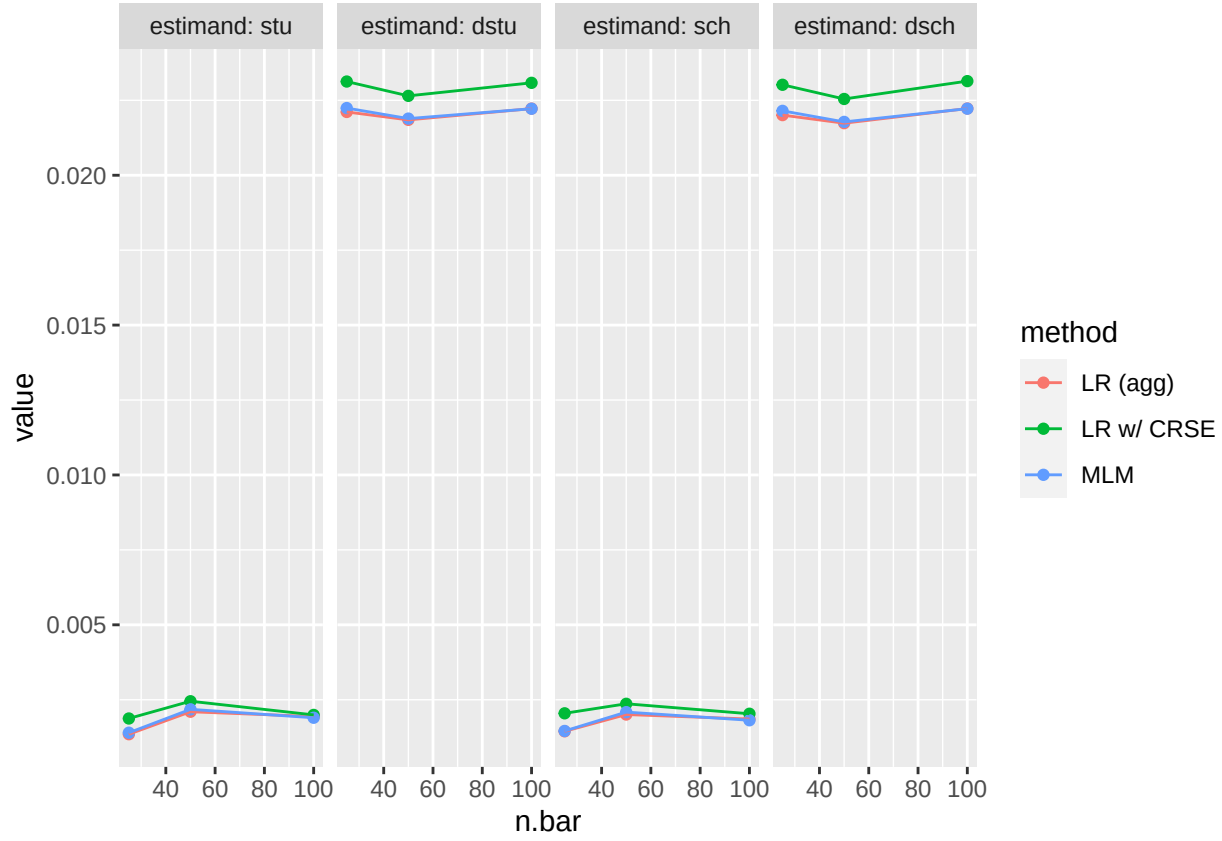
Not much variation still. Sample sizes don't matter!

See if faceting by sample sizes does anything? Nope. Pretty flat lines.

```
sruns %>%
  filter(method %in% key_methods) %>%
  # filter(size.impact.cor.district == 0) %>%
  filter(size.impact.cor.school == 0) %>%
  group_by(method, n.bar) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=n.bar, y=value, color=method)) +
  geom_point() +
  geom_line() +
```

```
  facet_grid( ~ estimand, labeller = "label_both")
```

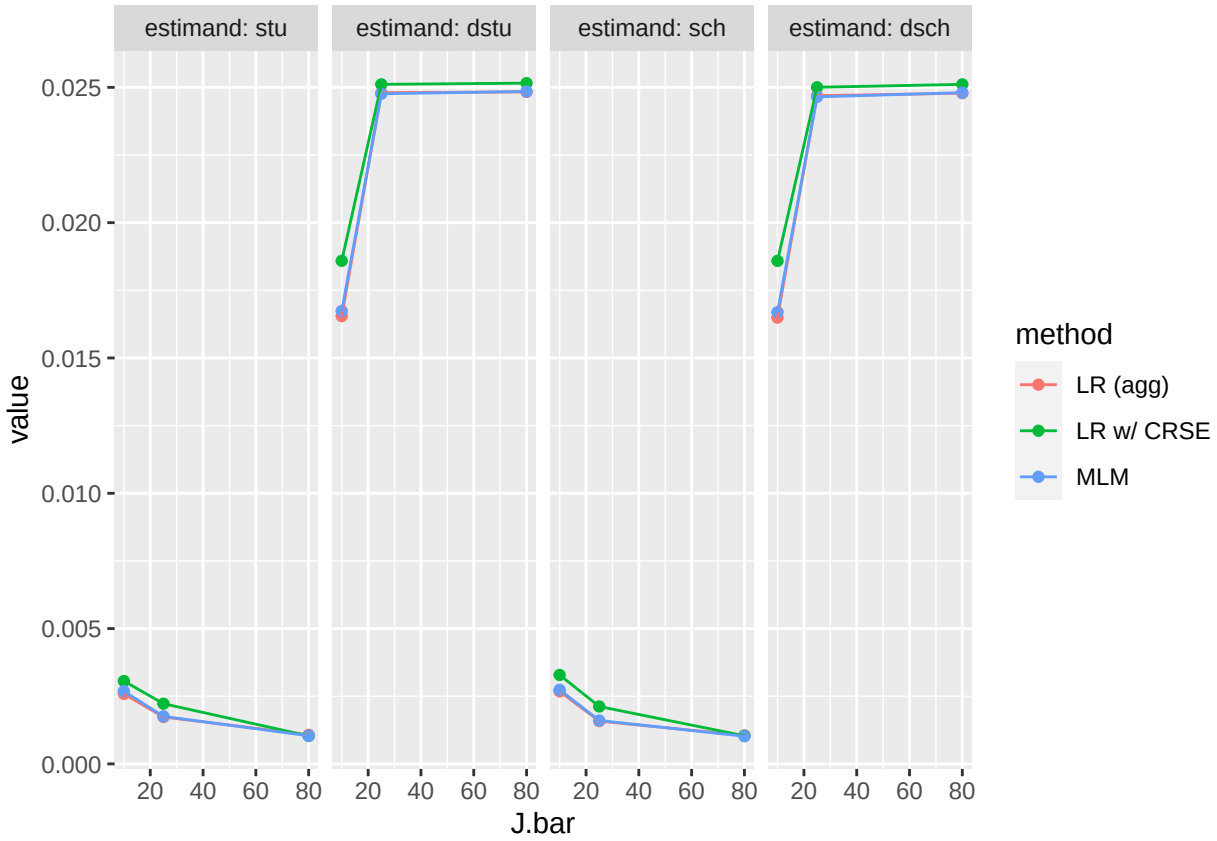
```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```



Seems to be a jump; once J increases, district-weighted estimands get worse?

```
sruns %>%
  filter(method %in% key_methods) %>%
  # filter(size.impact.cor.district == 0) %>%
  filter(size.impact.cor.school == 0) %>%
  group_by(method, J.bar) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=J.bar, y=value, color=method)) +
  geom_point() +
  geom_line() +
  facet_grid( ~ estimand, labeller = "label_both")
```

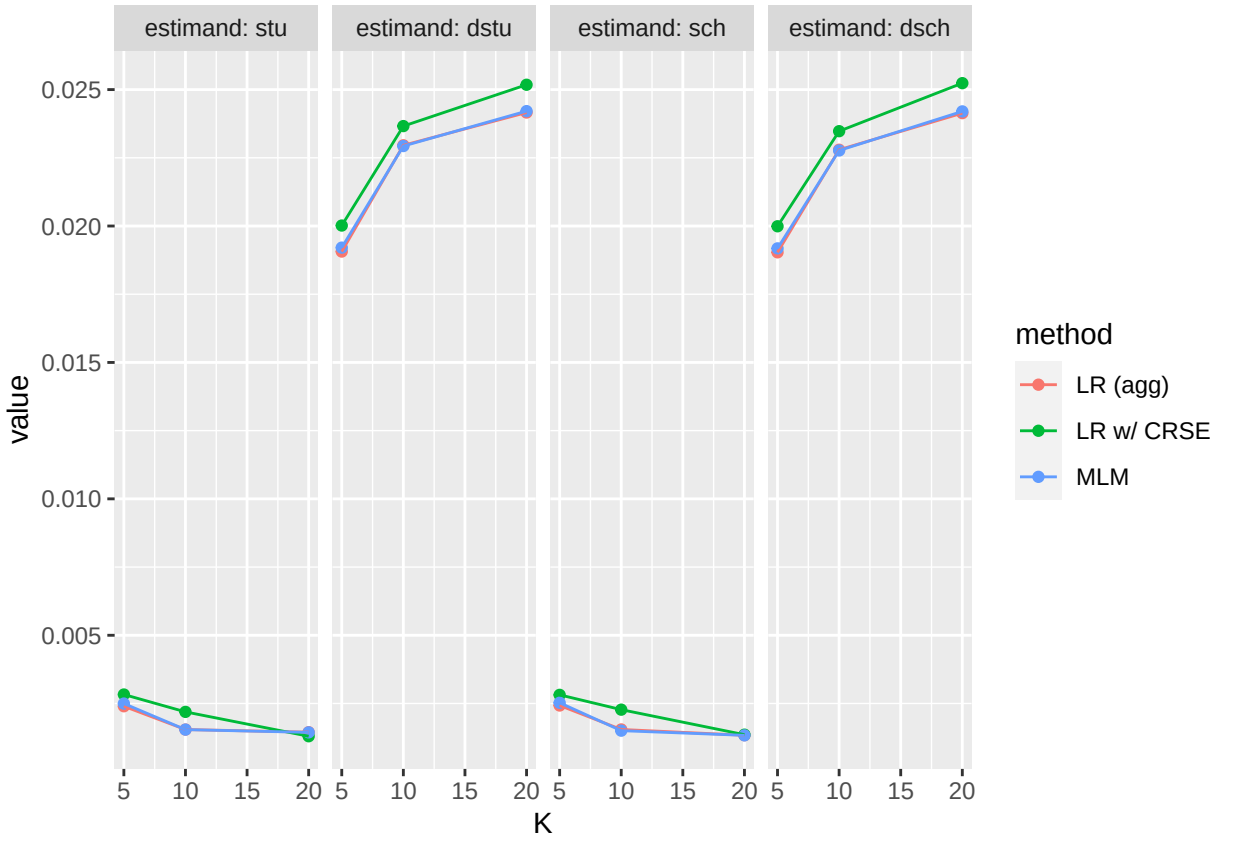
```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```



Now changing K… again, not much happens. Greater K might make district-weighted results worse.

```
sruns %>%
  filter(method %in% key_methods) %>%
  # filter(size.impact.cor.district == 0) %>%
  filter(size.impact.cor.school == 0) %>%
  group_by(method, K) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=K, y=value, color=method)) +
  geom_point() +
  geom_line() +
  facet_grid( ~ estimand, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```



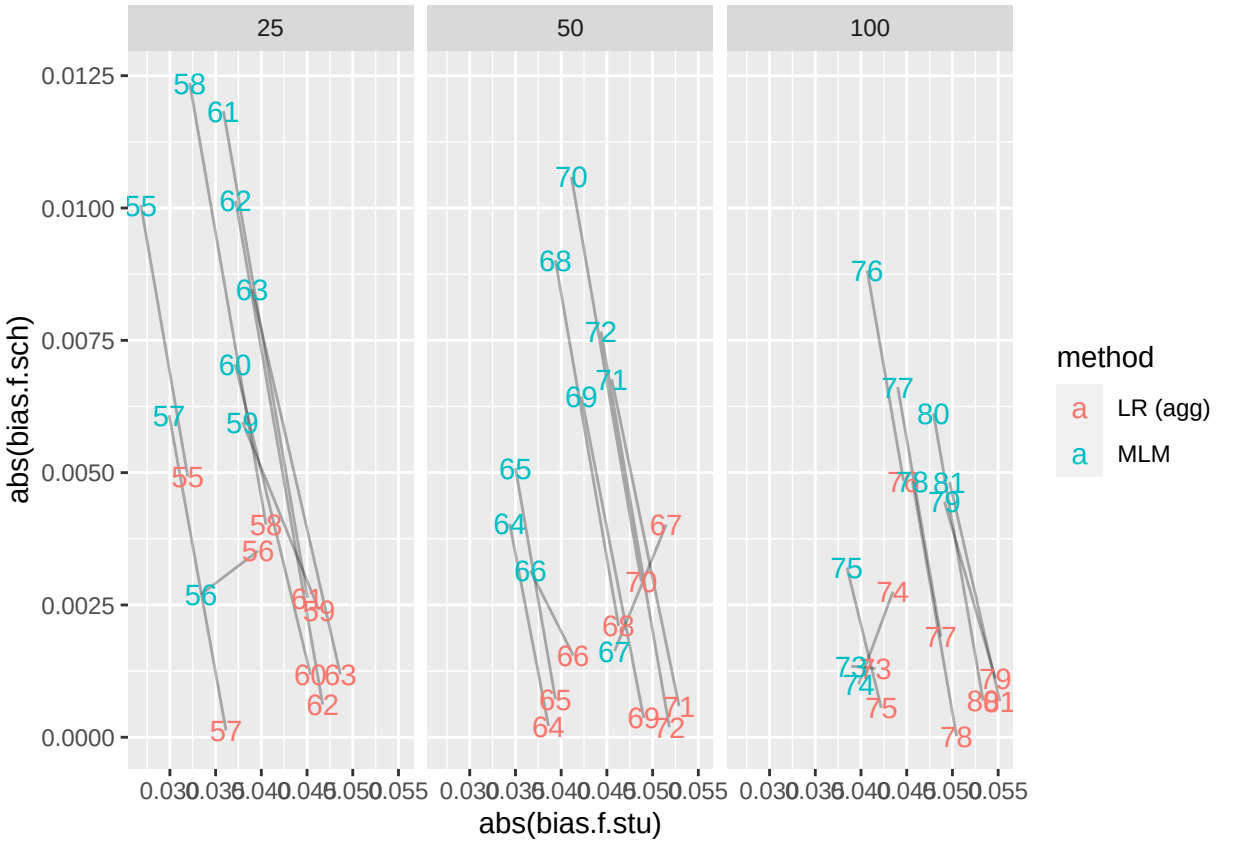
Conclusion: * when there is only school-effect correlation: maybe increasing n.bar and J makes estimators worse for their non-target estimand? * when there is only district-effect correlation: maybe increasing J and K makes estimators worse for the district-weighted estimands?

Unclear overall, not too much effect.

Exploring MLMs again

MLM is pretty consistently worse at sch (which it's good at nonetheless), and better at stu (which it's bad at regardless)

```
sruns %>%
  filter(method %in% c("LR (agg)", "MLM"),
         size.impact.cor.district == 0,
         size.impact.cor.school == 1) %>%
  ggplot(aes(x=abs(bias.f.stu), y=abs(bias.f.sch), color=method, label=ID)) +
  geom_line(aes(group=ID), color="black", alpha=0.3) +
  geom_text() +
  facet_wrap(~n.bar)
```



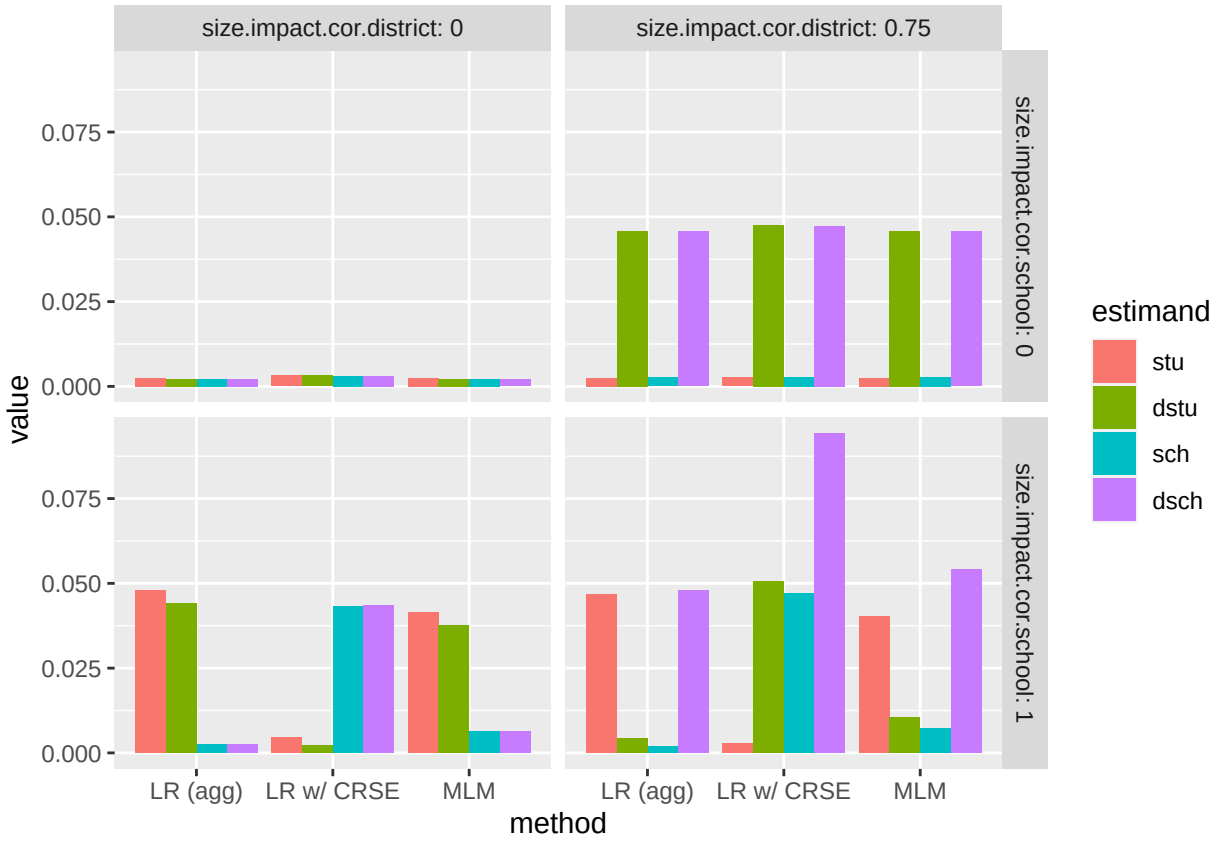
Across almost all settings, MLM just has slightly less student bias, and slightly more school bias. There's no real patterns to see...?

Messing with ICC values

```
load( "~/R/clusterRCTsim/results/aggregated_simulation_results_icc.RData" )
```

```
sruns %>%
  filter(method %in% key_methods) %>%
  group_by(method, size.impact.cor.school, size.impact.cor.district) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=method, y=value, fill=estimand)) +
  geom_col(position="dodge") +
  facet_grid(size.impact.cor.school ~ size.impact.cor.district, labeller = "label_both")
```

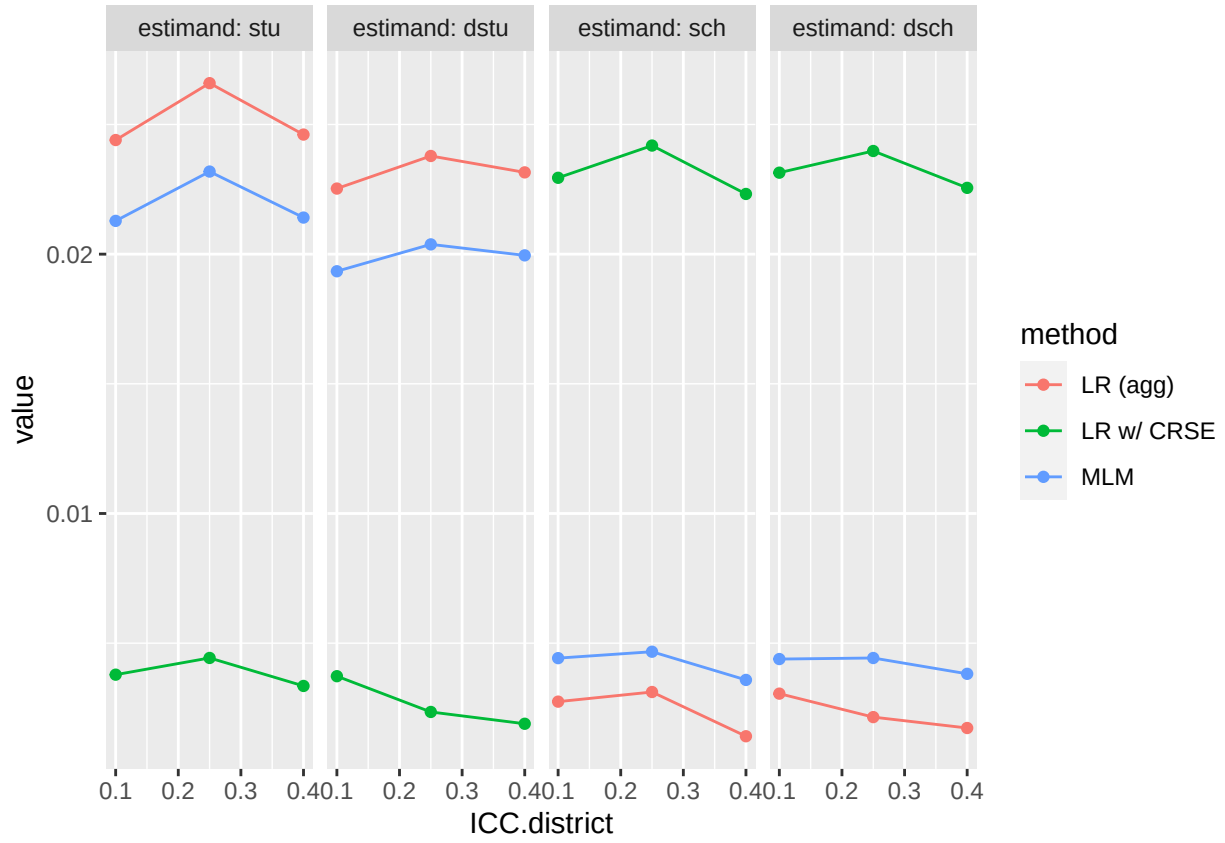
```
## `summarise()` has grouped output by 'method', 'size.impact.cor.school'. You can
## override using the `.groups` argument.
```



Increasing ICC.school maybe makes things worse for MLMs?

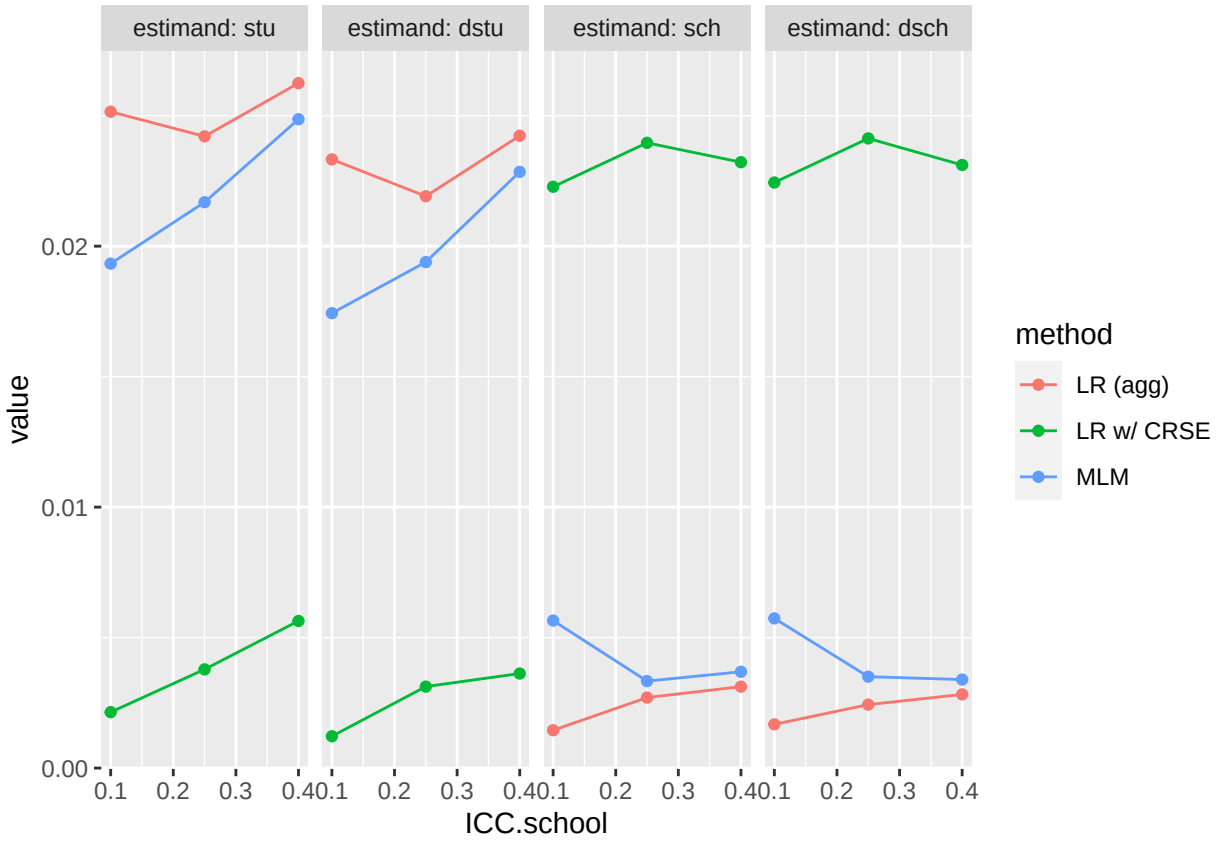
```
sruns %>%
  filter(method %in% key_methods) %>%
  filter(size.impact.cor.district == 0) %>%
  group_by(method, ICC.district) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=ICC.district, y=value, color=method)) +
  geom_point() +
  geom_line() +
  facet_grid( ~ estimand, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```




```
sruns %>%
  filter(method %in% key_methods) %>%
  filter(size.impact.cor.district == 0) %>%
  group_by(method, ICC.school) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=ICC.school, y=value, color=method)) +
  geom_point() +
  geom_line() +
  facet_grid( ~ estimand, labeller = "label_both")
```

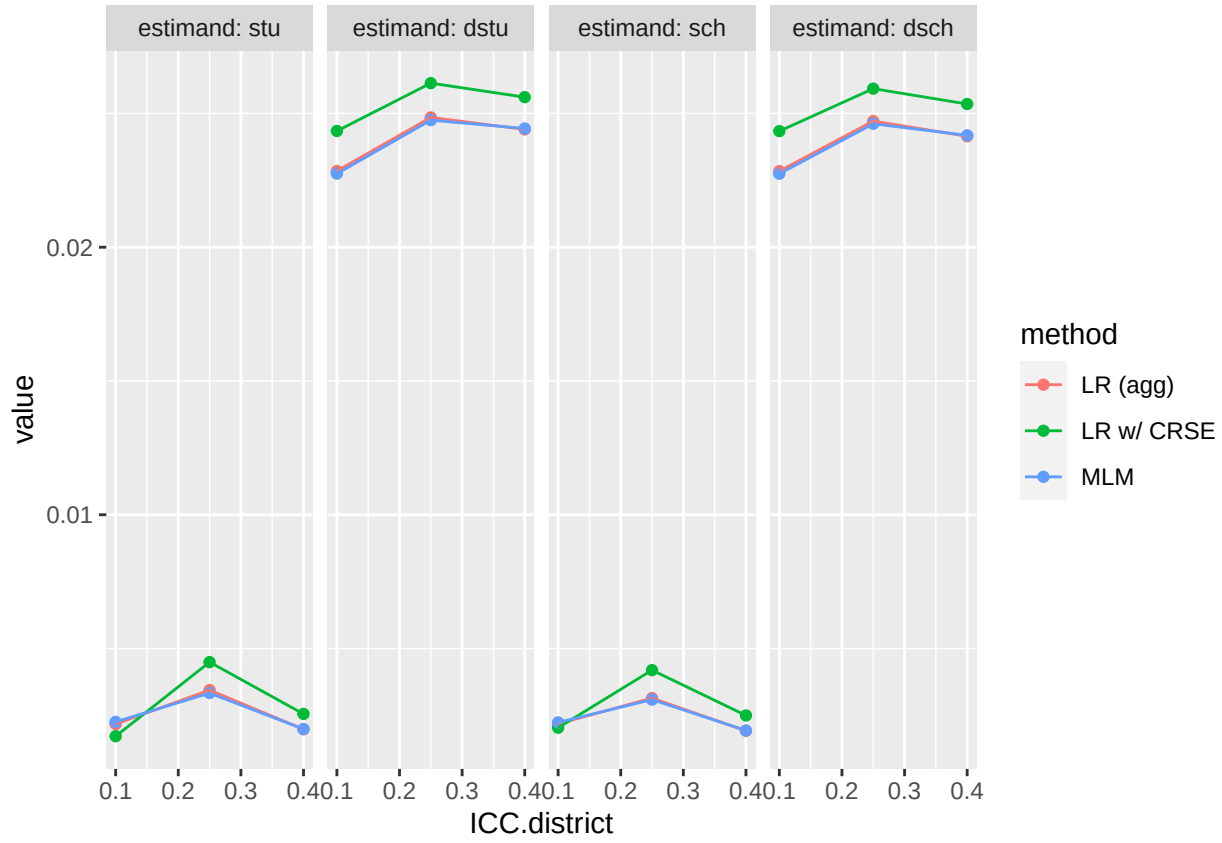
```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```



No effect below, when only district-level cor varies.

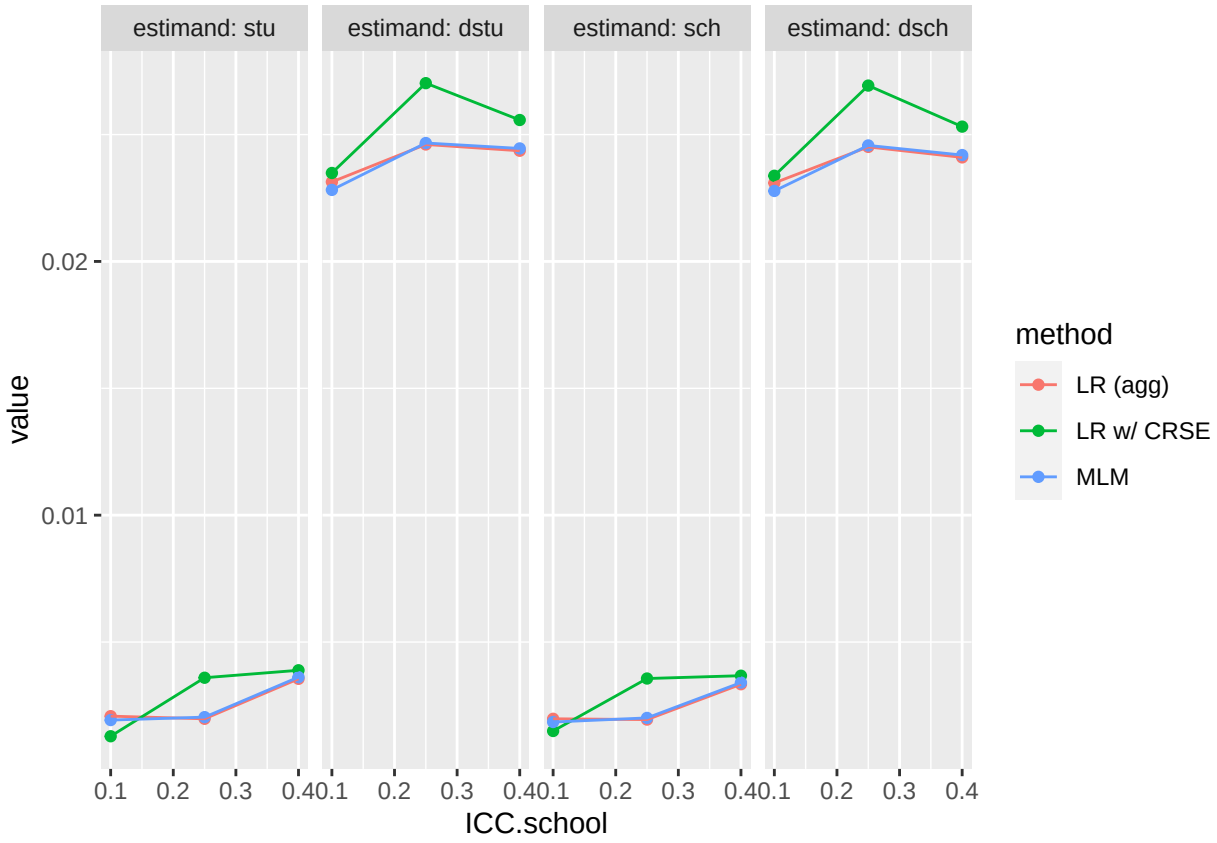
```
sruns %>%
  filter(method %in% key_methods) %>%
  filter(size.impact.cor.school == 0) %>%
  group_by(method, ICC.district) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=ICC.district, y=value, color=method)) +
  geom_point() +
  geom_line() +
  facet_grid( ~ estimand, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```




```
sruns %>%
  filter(method %in% key_methods) %>%
  filter(size.impact.cor.school == 0) %>%
  group_by(method, ICC.school) %>%
  summarize(across(starts_with("bias"), ~mean(abs(.)))) %>%
  pivot_longer(starts_with("bias")) %>%
  mutate(finite = str_detect(name, ".f"),
         estimand = as.factor(str_extract(name, "([^.]+$)"))) %>%
  mutate(estimand = factor(estimand, levels= c("stu", "dstu", "sch", "dsch"))) %>%
  filter(finite) %>%
  ggplot(aes(x=ICC.school, y=value, color=method)) +
  geom_point() +
  geom_line() +
  facet_grid( ~ estimand, labeller = "label_both")
```

```
## `summarise()` has grouped output by 'method'. You can override using the
## `.groups` argument.
```



Final takeaways

1. We have three methods: LR with CRSE, MLM, LR (agg). The first two target student-weighted estimands, the last targets school-weighted estimands
2. Correlating district impacts with district sizes leads to bias for all district-weighted estimands, i.e., none of our methods are good at targeting the district-weighted estimands
3. Correlating school impacts with school sizes leads to bias for non-target estimands (both district-weighted and non-district-weighted)
4. Sample sizes don't strongly affect results. Often increasing data makes estimators significantly more biased for the estimands they're not trying to target.
5. Not sure how to get MLMs to more closely target a given estimand. MLMs just generally have more school-estimand bias, less student-estimand bias than LR (agg).
6. Changing ICCs doesn't really affect things either?